

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285885

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

TOMINAGA; Ryoya

LIQUID STORAGE APPARATUS, SEMICONDUCTOR MANUFACTURING SYSTEM, AND LIQUID REPLENISHING METHOD

Abstract

A liquid storage apparatus includes: a container body storing a processing liquid for a substrate processing apparatus therein; a liquid level sensor provided in the container body, and detecting a liquid level of the processing liquid stored in the container body; a replenishment unit replenishing the processing liquid in the container body; and a control unit acquiring detection information from the liquid level sensor, and performing a processing. The liquid level sensor includes four or more detectors capable of detecting the liquid level of the processing liquid along a height direction of the container body, and the control unit sets at least three liquid level determination positions corresponding to different detectors, respectively, among the four or more detectors, and controls a replenishment of the processing liquid from the replenishment unit to the container body based on detection information of the set at least three liquid level determination positions.

Inventors: TOMINAGA; Ryoya (Iwate, JP)

Applicant: Tokyo Electron Limited (Tokyo, JP)

Family ID: 1000008508857

Appl. No.: 19/072373

Filed: March 06, 2025

Foreign Application Priority Data

JP 2024-037320

Mar. 11, 2024

Publication Classification

Int. Cl.: H01L21/67 (20060101); G05D9/12 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority from Japanese Patent Application No. 2024-037320, filed on Mar. 11, 2024, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a liquid storage apparatus, a semiconductor manufacturing system, and a liquid replenishing method.

BACKGROUND

[0003] Japanese Laid-Open Patent Publication No. 2003-146397 discloses a liquid storage apparatus (liquid supply apparatus) equipped with a liquid level sensor (detection unit) that detects the liquid level of a liquid stored in a container body (recharge tank). The liquid level sensor is configured to detect two liquid levels of a liquid, including the lowest liquid level and the highest liquid level.

[0004] Japanese Laid-Open Patent Publication No. 2008-522036 discloses a system, which stores a liquid (CVD precursor) in a container body (precursor storage tank) and supplies the vapor of the liquid to a semiconductor manufacturing apparatus while detecting the liquid level of the liquid using a liquid level sensor (level detection unit). The liquid level sensor is also configured to detect liquid levels of two set points (first and second set points).

SUMMARY

[0005] According to an aspect of the present disclosure, a liquid storage apparatus includes: a container body storing a processing liquid for a substrate processing apparatus therein; a liquid level sensor provided in the container body, and detecting a liquid level of the processing liquid stored in the container body; a replenishment unit replenishing the processing liquid in the container body; and a control unit acquiring detection information from the liquid level sensor, and performing a processing. The liquid level sensor includes four or more detectors capable of detecting the liquid level of the processing liquid along a height direction of the container body, and the control unit sets at least three liquid level determination positions corresponding to different detectors, respectively, among the four or more detectors, and controls a replenishment of the processing liquid from the replenishment unit to the container body based on detection information of the set at least three liquid level determination positions.

[0006] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a view illustrating an example of a semiconductor manufacturing system including a liquid storage apparatus according to a first embodiment.

[0008] FIG. 2A is a graph illustrating a relationship between the volume of a liquid stored in a container body and an output value of a liquid level sensor. FIG. 2B is a graph illustrating the shift

of a liquid level determination position set for each detector.

[0009] FIG. 3A is a cross-sectional view illustrating a state where a level of the container body is lowered. FIG. 3B is a cross-sectional view illustrating a state where the container body is replenished with a liquid.

[0010] FIG. 4 is a block diagram illustrating an example of functional blocks provided in a controller.

[0011] FIG. 5 is a flowchart illustrating a liquid supplying method and a liquid replenishing method in a semiconductor manufacturing system.

[0012] FIG. 6 is a flowchart illustrating a position correction method for correcting a liquid level determination position.

[0013] FIG. 7 is a view illustrating an example of a semiconductor manufacturing system 1 including a liquid storage apparatus 10 according to a second embodiment.

DETAILED DESCRIPTION

[0014] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made without departing from the spirit or scope of the subject matter presented here.

[0015] Hereinafter, embodiments for implementing the present disclosure are described with reference to the drawings. In each drawing, the same components are denoted by the same reference numeral, and overlapping descriptions thereof may be omitted.

First Embodiment

[0016] FIG. 1 is a view illustrating an example of a semiconductor manufacturing system 1 including a liquid storage apparatus 10 according to a first embodiment. As illustrated in FIG. 1, the semiconductor manufacturing system 1 is configured such that a processing liquid is supplied from the liquid storage apparatus 10 to a substrate processing apparatus 50, and a substrate processing is performed on a substrate W of the substrate processing apparatus 50 using the processing liquid. The type of substrate processing performed by the substrate processing apparatus 50 is not particularly limited, and includes, for example, a deposition process, an etching, a modification process, a stripping process, a rinsing process, and a cleaning process.

[0017] The processing liquid may be appropriately selected according to, for example, the contents of the substrate processing and the film type of the substrate W to be processed. For example, when the substrate processing is a cleaning process to clean the surface of the substrate W, a chemical liquid for cleaning the substrate W or pure water for rinsing the surface of the substrate W is applied as the processing liquid. In this case, the chemical liquid may be, for example, SPM, which is a mixture of sulfuric acid and hydrogen peroxide, APM, which is a mixture of ammonia and hydrogen peroxide, DHF, which is a mixture of hydrofluoric acid and pure water, or HPM, which is a mixture of hydrochloric acid and hydrogen peroxide. The pure water may be, for example, ultrapure water such as de-ionized water (DIW) from which ionic components have been removed. In the descriptions herein below, the processing liquid may be referred to simply as a liquid.

[0018] In the first embodiment, descriptions are made on an example of the semiconductor manufacturing system 1 that performs the cleaning process as the substrate processing. For example, the semiconductor manufacturing system 1 includes the liquid storage apparatus 10 that stores the cleaning liquid, a supply network 30 that supplies the liquid of the liquid storage apparatus 10, and the substrate processing apparatus 50 that actually performs the substrate processing. Further, the semiconductor manufacturing system 1 includes a controller 90 that manages the entire system and controls each component of the system.

[0019] The liquid storage apparatus 10 is configured to temporarily store an appropriate amount of liquid, and supply (or discharge) the liquid. For example, the liquid storage apparatus 10 includes a container body 11, an inlet port 12 that allows the flow of the liquid into the container body 11, an

outlet port **13** that allows the flow of the liquid out of the container body **11**, and an opening/closing valve **14** provided at the outlet port **13** (or a supply path **31** connected to the outlet port **13**).

[0020] Further, the liquid storage apparatus **10** includes a replenishment unit **15** that replenishes the liquid in the container body **11**, at the upstream side relative to the container body **11**. The replenishment unit **15** includes, for example, a replenishment source **16** of the liquid, an inflow path **17**, and an opening/closing valve **18**, which are provided separately from the container body **11**. The replenishment source **16** may be, for example, a mother tank of a liquid supplied from another supply source, or a liquid generating apparatus that generates a liquid. The liquid generating apparatus may be any of various apparatuses that, for example, distill a liquid, mix (add) an appropriate substance with (to) a liquid, filter a liquid, or extract a liquid. The opening/closing valve **18** opens and closes the flow channel of the inflow path **17** based on the control by the controller **90**. The replenishment unit **15** discharges the liquid from the replenishment source **16** to the inflow path **17** in a state where the opening/closing valve **18** is opened, and replenishes the container body **11** with the liquid through the inlet port **12**.

[0021] The container body **11** is formed, for example, in the shape of an inverted conical trapezoid tapered from the ceiling to the bottom thereof. A storage space **11s** of the liquid is formed inside the container body **11** to conform to the shape of the container body **11**. The size (volume) of the storage space **11s** may be set according to, for example, the amount of liquid necessary for the cleaning process. The shape of the container body **11** is not particularly limited, and may be, for example, a cylindrical shape or a rectangular cylindrical shape having a constant cross-sectional area in the vertical direction.

[0022] The inlet port **12** is provided at the ceiling of the container body **11**, and the outlet port **13** is provided at the bottom of the container body **11**. The inlet port **12** is connected to the inflow path **17** of the replenishment unit **15**, and protrudes slightly from the ceiling into the storage space **11s**. The inlet port **12** allows the liquid of the replenishment unit **15** to flow into the storage space **11s** from the inlet at the lower end thereof. Further, in order to reduce, for example, the turbulence of the liquid in the storage space **11s**, for example, the inlet port **12** may protrude even to the vicinity of the bottom of the container body **11**, and discharge the liquid from the inlet at the lower end thereof. Meanwhile, the outlet port **13** is provided substantially at the center of the bottom of the container body **11**. The liquid storage apparatus **10** discharges the liquid to the supply network **30**, using the weight of the liquid itself (pressure) stored in the storage space **11s**. Without being limited to this configuration, the semiconductor manufacturing system **1** may be configured such that a pump or the like is provided in the supply network **30** to pressure-feed the liquid of the container body **11**.

[0023] Further, the liquid storage apparatus **10** according to the first embodiment includes a heating unit **19** that heats the liquid stored in the container body **11**. In the illustrated example, the heating unit **19** is provided inside the peripheral wall of the container body **11**. However, the installation location of the heating unit **19** is not limited thereto, and the heating unit **19** may be provided at the bottom, the ceiling, or the outside of the container body **11**. The heating unit **19** is connected to the controller **90** via a driver (not illustrated), and heats the liquid of the container body **11** to a target temperature in the manner that a supplied power in the driver is controlled according to the target temperature instructed by the controller **90**. The liquid storage apparatus **10** may include a temperature sensor (not illustrated) in the container body **11** to provide a feedback on the actual temperature of the liquid. The configuration of the heating unit **19** is not particularly limited, and for example, an electric heating wire provided inside the container body **11** or a sheet heater covering the outside of the container body **11** may be applied.

[0024] Further, the liquid storage apparatus **10** includes a liquid level sensor **20** that detects the liquid level of the liquid stored in the storage space **11s** of the container body **11**. The configuration of the liquid level sensor **20** is described in detail herein later.

[0025] The supply network **30** includes a supply path **31** of the liquid, an opening/closing valve **32**, and a flow rate controller **33**. In the supply path **31** of the supply network **30**, for example, branching sections or merging sections are formed according to the configuration of the substrate processing apparatus **50**. For example, in the example of FIG. **1**, the substrate processing apparatus **50** includes a plurality of nozzles **52**, and the supply path **31** branches into a plurality of branch paths **311** and **312** from midway positions thereof to supply appropriate liquids to the plurality of nozzles **52**, respectively. The supply network **30** includes the opening/closing valve **32** and the flow rate controller **33** in each of the plurality of branch paths **311** and **312**.

[0026] Further, the supply network **30** includes gas supply units **40** in the plurality of branch paths **311** and **312**, respectively, to supply an inert gas such as nitrogen (N.sub.2) gas to the substrate processing apparatus **50**. The gas supply units **40** include gas supply paths **41** connected to the plurality of branch paths **311** and **312**, respectively. Further, each gas supply unit **40** is provided with, for example, a gas tank **42** storing the inert gas, an opening/closing valve **43**, and a flow rate controller **44** on each gas supply path **41**. The gas supply units **40** supply the inert gas to the substrate processing apparatus **50** through the gas supply paths **41** and the branch paths **311** and **312**, in order to purge residual liquid in the flow paths of the branch paths **311** and **312** or in the substrate processing apparatus **50**.

[0027] Without being limited to the configuration where the supply network **30** supplies the liquid from one liquid storage apparatus **10** to one substrate processing apparatus **50**, the supply network **30** may be configured to supply the liquid from a plurality of liquid storage apparatuses **10** to one substrate processing apparatus **50**. FIG. **1** illustrates an example of a configuration where a first liquid (e.g., SPM) is supplied from a first liquid storage apparatus **10A**, and a second liquid (e.g., DIW) is supplied from a second liquid storage apparatus **10B**. Further, the supply network **30** may be configured to supply the liquid from one liquid storage apparatus **10** to a plurality of substrate processing apparatuses **50**.

[0028] The substrate processing apparatus **50** includes a processing container **51** that accommodates the substrate **W**, a plurality of nozzles **52** provided inside the processing container **51**, and a substrate support unit **53** that supports the substrate **W** in the processing container **51**.

[0029] Each nozzle **52** includes a protruding portion that extends vertically in the processing container **51**, and ejects the liquid from an outlet **520** of the protruding portion toward the substrate **W**. Further, the nozzles **52** are movable in the processing container **51** by a moving mechanism (not illustrated). For example, each nozzle **52** is configured to move between an ejection position at the center of the substrate support unit **53** (substrate **W**) and a standby position radially outward from the substrate support unit **53**, based on the control by the controller **90**.

[0030] The plurality of nozzles **52** are connected to the branch paths **311** and **312**, respectively, leading to the first liquid storage apparatus **10A**, and connected to the supply path **31** leading to the second liquid storage apparatus **10B**. Thus, each nozzle **52** may eject the liquid supplied from the paths connected thereto, from the outlet **520** to supply the liquid to the surface of the substrate **W**.

[0031] The substrate support unit **53** places the substrate **W** on the upper surface thereof. In order to hold the substrate **W** placed on the upper surface, the substrate support unit **53** may include securing means such as electrostatic adsorption, suction adsorption, and a mechanical lock. The substrate support unit **53** may be configured to be rotatable around the center of the upper surface thereof in the processing container **51** by a rotation mechanism (not illustrated). For example, the substrate processing apparatus **50** supplies the liquid from the nozzles **52** positioned at the center of or above the substrate **W** while rotating the substrate **W** by the substrate support unit **53**. Thus, the liquid ejected onto the substrate **W** spreads across the substrate **W** outwardly in the radial direction while wetting the substrate **W** due to the centrifugal force of the substrate **W**, so that the entire surface of the substrate **W** may be immersed with the liquid.

[0032] The controller **90** of the semiconductor manufacturing system **1** may be a computer including, for example, a processor, a memory, an input/output interface, and a communication

interface (not illustrated). The processor may be any one of, for example, a central processing unit (CPU), a graphics processing unit (GPU), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and a circuit including a plurality of discrete semiconductors, or an electronic circuit, which is a combination thereof, and executes and processes programs stored in the memory. The memory includes a main storage device including, for example, a semiconductor memory, and an auxiliary storage device including, for example, a disk, a drive, and a semiconductor memory (flash memory).

[0033] The controller **90** of the semiconductor manufacturing system **1** controls the operations of the liquid storage apparatus **10**, the supply network **30**, and the substrate processing apparatus **50** to supply the liquid and perform the substrate processing on the substrate **W**. At this time, the controller **90** acquires detection information from the liquid level sensor **20** of the liquid storage apparatus **10** to manage the liquid stored in the container body **11** and perform, for example, the discharge or inflow (replenishment) of the liquid. In the present embodiment, descriptions are made on an example where the controller **90** of the semiconductor manufacturing system **1** controls each component of the liquid storage apparatus **10**. However, the semiconductor manufacturing system **1** may include a control unit (not illustrated) dedicated for the liquid storage apparatus **10**, and control each component of the liquid storage apparatus **10** by the control unit receiving an instruction from the controller **90**.

[0034] The liquid level sensor **20** of the liquid storage apparatus **10** is provided in the storage space **11s** of the container body **11**, and extends in a straight line shape along the height direction (vertical direction). The liquid level sensor **20** includes a plurality of detectors **21** (e.g., **25** detectors) at different positions along the height direction, and detects the presence or absence of the liquid level of the liquid at each detector **21**. As for the liquid level sensor **20**, a float-type level sensor may be applied, in which floats are arranged as the detectors **21** at equal intervals from each other with respect to a frame **22** extending in the height direction. For example, the detectors **21** are configured with, for example, magnets, and open and close a magnet switch by floating up due to the buoyancy of the liquid and descending when there is no liquid. The liquid level sensor **20** configured as described above may detect the liquid level in the container body **11** in steps according to the number of detectors **21**. The type of the liquid level sensor **20** is not limited to the float-type liquid level sensor, but, for example, an optical type, a capacitance type, an electrode type, or a pressure (differential pressure) type may be applied.

[0035] FIG. 2A is a graph illustrating a relationship between the volume of the liquid stored in the container body **11** and an output value of the liquid level sensor **20**. FIG. 2B is a graph illustrating the shift of a liquid level determination position set for each detector **21**. Since the detectors **21** of the liquid level sensor **20** are arranged at equal intervals along the height direction, the liquid level sensor **20** may output an output value proportional in steps with respect to the volume of the liquid in the container body **11**, as illustrated in FIG. 2A. The controller **90** may have information, in advance, in which the output value of the liquid level sensor **20** is associated with the volume of the liquid (the amount of liquid) (e.g., map information), and may recognize the volume of the liquid in the container body **11** based on the output value of the liquid level sensor **20** during the operation.

[0036] The number of detectors **21** of the liquid level sensor **20** is not limited to **25**, and may be less or more than **25**. However, it is assumed that when the number of liquid level determination positions to be described herein later is “**n**,” the liquid level sensor **20** includes at least an **n**+1 number of detectors **21**. In the first embodiment, three liquid level determination positions are set. Thus, the liquid level sensor **20** may include four or more detectors **21** along the height direction of the container body **11**.

[0037] As illustrated in FIGS. **1** and **2A**, the controller **90** sets at least three liquid level determination positions for the plurality of detectors **21** of the liquid level sensor **20** (**25** detectors), and detects the presence or absence of the liquid level at the set liquid level determination positions. The three liquid level determination positions include a highest position **HP**, a middle

position MP, and a lowest position LP in an order from the vertical upper side to the vertical lower side.

[0038] The highest position HP is information indicating a storage limit position when the liquid is stored in the container body **11**. When the liquid is stored exceeding the highest position HP, the controller **90** stops the supply of the liquid to the container body **11** by reporting, for example, a warning or stopping the operation of the system.

[0039] The middle position MP is information indicating a determination threshold value for determining the replenishment of the liquid by the replenishment unit **15**. Hereinafter, the control based on the middle position MP, which is the liquid level determination position, is described with reference to FIGS. **3A** and **3B**. FIG. **3A** is a view illustrating a state where the liquid level of the liquid in the container body **11** becomes lower than the middle position MP. FIG. **3B** is a view illustrating a state where the container body **11** is replenished with the liquid from the replenishment unit **15**.

[0040] As the liquid storage apparatus **10** supplies the liquid from the container body **11** to the substrate processing apparatus **50** during the substrate processing, the liquid level of the liquid in the container body **11** is lowered. The controller **90** regularly detects the liquid level of the liquid in the container body **11** by the liquid level sensor **20**, and when it is determined that the liquid level becomes lower than the middle position MP, the controller **90** starts replenishing the liquid in the container body **11** from the replenishment unit **15**. For example, when the replenishment of the liquid in a specific amount from the replenishment unit **15** is repeated, and it is determined that the liquid level exceeds the middle position MP, the liquid storage apparatus **10** stops the replenishment of the liquid by the replenishment unit **15**. Alternatively, the liquid storage apparatus **10** may replenish the container body **11** with the liquid in an amount sufficiently exceeding the middle position MP (a specific amount of liquid) all at once from the replenishment unit **15**.

[0041] The lowest position LP is information indicating a supply limit position when the liquid is supplied from the container body **11** to the substrate processing apparatus **50**. When the liquid is discharged to become lower than the lowest position LP, the controller **90** stops the supply of the liquid from the container body **11** by reporting, for example, a warning or stopping the operation of the system. The lowest position LP may be set to a position with a certain height from the bottom of the container body **11**, to ensure the amount of liquid used for one substrate processing. Thus, without immediately stopping the substrate processing when the liquid falls below the lowest position LP, the controller **90** may stop the supply of the liquid after the substrate processing is completed.

[0042] Further, without being limited to setting the three liquid level determination positions, the controller **90** may set four or more liquid level determination positions. For example, the controller **90** may set two positions (a replenishment start position for determining the start of the replenishment of liquid by the replenishment unit **15** and a replenishment stop position for determining the stop of the replenishment of liquid by the replenishment unit **15**) as middle positions MP, in addition to the highest position HP and the lowest position LP,

[0043] FIG. **4** is a block diagram illustrating functional blocks of the controller **90**. The controller **90** causes the processor to execute the programs stored in the memory, to form functional units that process the detection information of the liquid level sensor **20** of the liquid storage apparatus **10**, as illustrated in FIG. **4**. For example, the controller **90** is provided with, therein, a liquid level determination position setting unit **101**, a detection information acquisition unit **102**, a liquid level determination unit **103**, a control processing unit **104**, and a liquid state estimation process unit **105**.

[0044] The liquid level determination position setting unit **101** sets the liquid level determination positions of the liquid in the container body **11**, automatically by the controller **90** or by a user of the semiconductor manufacturing system **1**. For example, the liquid level determination position setting unit **101** sets each liquid level determination position by displaying setting screen

information for setting the liquid level on a monitor (not illustrated) or the like of the controller **90**, and allowing the user to operate the setting screen information.

[0045] Alternatively, the liquid level determination position setting unit **101** may adjust the liquid level determination positions (e.g., the middle position MP) based on the contents of the substrate processing of the substrate processing apparatus **50**. The liquid level determination position setting unit **101** calculates the consumption of the liquid used for the substrate processing according to, for example, a recipe for the substrate processing stored in the memory, and sets the storage amount of liquid to be stored in the container body **11** during the substrate processing based on the calculated consumption of the liquid, that is, the middle position MP. For example, the liquid level determination position setting unit **101** may raise the middle position MP when the consumption of the liquid is large, and lower the middle position MP when the consumption of the liquid is small.

[0046] The detection information acquisition unit **102** acquires the detection information of the liquid level sensor **20** periodically or at appropriate timings, and stores the acquired detection information in the memory. In the memory, the information on the liquid level from the liquid level sensor **20** may be stored in association with information such as detection timings.

[0047] The liquid level determination unit **103** compares the acquired detection information of the liquid level (actual liquid level of the liquid stored in the container body **11**) with each preset liquid level determination position, to determine the state of the liquid level. For example, the liquid level determination unit **103** determines whether the actual liquid level is equal to or higher than, or lower than the middle position MP.

[0048] The control processing unit **104** controls the operation of the liquid storage apparatus **10** based on, for example, the status of the substrate processing in the substrate processing apparatus **50** and the result of the determination by the liquid level determination unit **103**. For example, the control processing unit **104** opens the opening/closing valve **14** with the start of the substrate processing to supply the liquid to the substrate processing apparatus **50**, and closes the opening/closing valve **14** with the end of the substrate processing to stop the supply of the liquid to the substrate processing apparatus **50**.

[0049] When the actual liquid level becomes lower than the middle position MP, the control processing unit **104** replenishes the liquid in the container body **11** from the replenishment unit **15**. The replenishment of the liquid from the replenishment unit **15** may be performed immediately during the substrate processing, or may be performed after waiting for the end of the substrate processing. For example, when the temperature of the liquid is being adjusted by the heating unit **19**, the liquid storage apparatus **10** sets the timing for supplying the liquid to a timing after the end of the substrate processing. Thus, the temperature of the liquid in the container body **11** may be suppressed from varying during the substrate processing due to the replenishment.

[0050] The liquid state estimation process unit **105** is a functional unit that automatically performs, for example, the adjustment of the liquid level determination position, the adjustment of a replenishment amount of the liquid replenished by the replenishment unit **15**, and the request for replacing the container body **11** according to various statuses of the liquid storage apparatus **10** or the substrate processing apparatus **50**. The statuses requiring the adjustment of the liquid level determination position include, for example, statuses (1) and (2) below.

[0051] (1) A thermal expansion occurs in the liquid according to temperatures.

[0052] (2) A precipitate is deposited in the container body **11**.

[0053] Status (1) is, for example, a status in which the liquid in the container body **11** is heated (adjusted to the target temperature) by the heating unit **19**. The liquid has its inherent thermal expansion coefficient, and expands according to the amount of heat received from the heating unit **19** and the thermal expansion coefficient. Thus, the liquid state estimation process unit **105** performs a process of optimizing (adjusting) the liquid level determination position according to the target temperature of the container body **11** (the heating unit **19**), using the thermal expansion coefficient of the liquid as a factor. For example, when the liquid of the container body **11** expands

so that the detector **21** corresponding to the middle position MP deviates, the liquid state estimation process unit **105** performs a process of shifting the detector **21** of the middle position MP to another detector **21** according to the expansion of the liquid. In this case, as illustrated in FIG. 2B, the liquid state estimation process unit **105** may shift the highest position HP and/or the lowest position LP, in addition to the middle position MP, in an integrated manner as illustrated in FIG. 2B.

[0054] Status (2) is, for example, a status in which a precipitate is deposited in the container body **11** as the substrate processing is performed multiple times, and the replenishment of the liquid in the container body **11** is repeated, when impurities or the like that easily precipitate are mixed in the liquid. The amount of precipitate may be estimated by monitoring the cumulative replenishment amount of liquid supplied to the container body **11** and/or the cumulative consumption of the liquid used in the substrate processing. For example, the liquid state estimation process unit **105** continuously stores the cumulative replenishment amount and the cumulative consumption to estimate the liquid level of the container body **11**, and estimate the amount of precipitate based on the state of deviation between the estimated liquid level and the actual liquid level (current liquid level). Alternatively, the liquid state estimation process unit **105** may perform, for example, experiments or simulations to obtain a function or map information, which associates the cumulative replenishment amount of liquid and the amount of precipitate with each other, and estimate the amount of precipitate from the cumulative replenishment amount.

[0055] The liquid state estimation process unit **105** may perform a process of shifting each liquid level determination position (the highest position HP, the middle position MP, and the lowest position LP), based on the estimated amount of precipitate. For example, when the corresponding position between each detector **21** of the liquid level sensor **20** and each liquid level determination position deviates by one step as the liquid level rises due to the precipitate, the liquid state estimation process unit **105** adjusts each liquid level determination position by one step. Alternatively, when the amount of precipitate in the container body **11** increases, the liquid state estimation process unit **105** may request the user to replace the container body **11** through a user interface such as the monitor of the controller **90**.

[0056] Further, the liquid state estimation process unit **105** may perform a process of monitoring the liquid level of the liquid by the liquid level sensor **20** during the replenishment of the liquid from the replenishment unit **15**, and changing the replenishment amount of liquid replenished by the replenishment unit **15**. For example, when replenishing the liquid, the current liquid level relative to a target liquid level may be identified from the detection information of each detector **21** of the liquid level sensor **20**, and therefore, the status of the liquid replenishment may be substantially estimated. Thus, the liquid state estimation process unit **105** may increase the replenishment amount in the initial stage of the liquid replenishment from the replenishment unit **15**, and decrease the replenishment amount when the liquid level reaches the vicinity of the target liquid level. As a result, the liquid storage apparatus **10** may reduce the time for the liquid replenishment, suppress, for example, the fluctuation of the liquid level after the replenishment, and improve the efficiency of the entire operation including the substrate processing.

[0057] The liquid storage apparatus **10** and the semiconductor manufacturing system **1** according to the embodiment are basically configured as described above, and hereinafter, the operations thereof are described with reference to FIGS. 5 and 6. FIG. 5 is a flowchart illustrating a liquid replenishing method. FIG. 6 is a flowchart illustrating a method of correcting the liquid level determination position.

[0058] The controller **90** of the semiconductor manufacturing system **1** controls steps **S101** to **S109** illustrated in FIG. 5, to perform the substrate processing by the substrate processing apparatus **50** and manage the liquid of the liquid storage apparatus **10**.

[0059] Specifically, prior to starting the substrate processing by the substrate processing apparatus **50**, the controller **90** sets the liquid level determination positions (the highest position HP, the middle position MP, and the lowest position LP) corresponding to different detectors **21**,

respectively, among the detectors **21** of the liquid level sensor **20** (step **S101**). For example, the liquid level determination position setting unit **101** of the controller **90** sets the reference liquid level determination positions based on the contents of the substrate processing as described above. [0060] Further, prior to starting the substrate processing, the liquid state estimation process unit **105** determines whether a correction start condition for performing a correction method to adjust the liquid level determination positions is satisfied (step **S102**). The correction start condition may include, for example, a case where the time period during which the substrate processing has been performed or the number of times of performing the substrate processing (e.g., the cumulative consumption of the liquid) is equal to or more than a preset threshold value, and a case where the liquid in the container body **11** is heated according to the substrate processing. When it is determined that the correction start condition is satisfied (step **S102**: YES), the controller **90** proceeds to the subroutine of the correction method. Meanwhile, when it is determined that the correction start condition is not satisfied (step **S102**: NO), the controller **90** does not perform the correction method at this time, and proceeds to step **S103**. The timing for determining the correction start condition is not limited to the timing prior to starting the substrate processing, but may be any appropriate timing such as the time when the liquid level of the container body **11** becomes lower than the middle position MP, after the end of the substrate processing. When the correction is performed at a timing other than the timing prior to starting the substrate processing, the controller **90** may store the corrected liquid level determination positions in the memory.

[0061] In step **S103**, the controller **90** controls the substrate processing apparatus **50** to perform the substrate processing, after performing, for example, the temperature adjustment of the liquid in the container body **11** according to the substrate processing (heating by the heating unit **19**), and the adjustment of the pressure or temperature of the substrate processing apparatus **50**. During the substrate processing, the liquid storage apparatus **10** supplies the liquid from the container body **11** to the substrate processing apparatus **50** through the supply network **30**. As the substrate processing is progressed, the liquid level of the liquid in the container body **11** is gradually lowered.

[0062] The controller **90** monitors the liquid level of the liquid in the container body **11** by acquiring the detection information of the liquid level from the liquid level sensor **20** during the substrate processing, to determine whether the liquid level is equal to or higher than the middle position MP (step **S104**). When it is determined that the liquid level is equal to or higher than the middle position MP (step **S104**: YES), the process proceeds to step **S105**. In step **S105**, the controller **90** determines whether to terminate the substrate processing. When it is determined to continue the substrate processing (step **S105**: NO), the process returns to step **S103** to repeat the same and subsequent steps. Meanwhile, when it is determined to terminate the substrate processing (step **S105**: YES), the process proceeds to step **S106** to perform termination processes such as stopping the liquid supply by the liquid storage apparatus **10** and taking the substrate W out of the substrate processing apparatus **50**, and terminate the substrate processing.

[0063] Meanwhile, when it is determined that the liquid level is lower than the middle position MP (step **S104**: NO), the controller **90** proceeds to step **S107**. In step **S107**, the controller **90** determines whether to terminate the substrate processing. When it is determined to continue the substrate processing (step **S107**: NO), the process returns to step **S103** to repeat the same and subsequent steps. Meanwhile, when it is determined to terminate the substrate processing (step **S107**: YES), the process proceeds to step **S108** to perform termination processes such as stopping the liquid supply by the liquid storage apparatus **10** and taking the substrate W out of the substrate processing apparatus **50**, and terminate the substrate processing.

[0064] Further, the controller **90** replenishes the container body **11** with the liquid from the replenishment unit **15**, based on the liquid level falling below the preset middle position MP (step **S109**). As a result, the container body **11** is replenished with the liquid exceeding the middle position MP, and the liquid storage apparatus **10** may stably supply the liquid for the next substrate processing.

[0065] When the correction start condition is satisfied in step S102 as described above, the controller **90** performs the subroutine of the correction method illustrated in FIG. **6** to correct the liquid level determination positions. For example, in the correction method, the liquid state estimation process unit **105** of the controller **90** first estimates the amount of precipitate being deposited in the container body **11** based on the cumulative replenishment amount, the cumulative consumption, and the actual liquid level as described above (step S201).

[0066] Next, the liquid state estimation process unit **105** determines whether the estimated amount of precipitate is varying by a specific value or more (step S202). In this case, the specific value is, for example, an amount at which the detectors **21** of the liquid level sensor **20** setting the liquid level determination positions needs to be shifted, and may be set in advance through experiments or simulations. The liquid state estimation process unit **105** proceeds to step S203 when it is determined that the amount of precipitate is varying by the specific value or more (step S202: YES), and proceeds to step S204, without performing step S203, when it is determined that the change of the amount of precipitate is less than the specific value (step S202: NO).

[0067] In step S203, the controller **90** changes the liquid level determination positions according to the estimated amount of precipitate. For example, when the detectors **21** deviate upwardly by only one step due to the amount of precipitate, the controller **90** performs the process of shifting each of the detectors **21** corresponding to the set liquid level determination positions (the highest position HP, the middle position MP, and the lowest position LP) upwardly by one step.

[0068] Further, the controller **90** determines whether the liquid undergoes a significant thermal expansion, when the temperature of the liquid in the container body **11** is changed through the heating by the heating unit **19** (step S204). For example, the controller **90** calculates the degree of expansion of the liquid when the temperature of the liquid is increased to the target temperature through the heating by the heating unit **19**, based on, for example, the thermal expansion coefficient of the liquid, the target temperature, and the volume or cross-sectional area of the container body **11**. The degree of expansion is an indicator of the change of the liquid along the height direction in the container body **11**, and the liquid rises as the degree of expansion is significant. The controller **90** determines that the thermal expansion is significant when the degree of expansion of the liquid affects the positions of the detectors **21** of the liquid level sensor **20** (step S204: YES), and proceeds to step S205. Meanwhile, when it is determined that the thermal expansion is not significant (step S204: NO), step S205 is not performed. Further, when the thermal expansion coefficient of the liquid varies due to the change of the liquid stored in the container body **11**, the controller **90** may also determine the thermal expansion of the liquid at this timing.

[0069] In step S205, the controller **90** changes the liquid level determination positions according to the degree of expansion of the liquid. For example, when the detectors **21** deviate upwardly by only one step due to the thermal expansion of the liquid, the controller **90** performs the process of shifting each of the detectors **21** corresponding to the set liquid level determination positions (the highest position HP, the middle position MP, and the lowest position LP) upwardly by one step. Further, the controller **90** may add the amount of change in height position of the liquid surface based on the amount of precipitate and the amount of change in height position of the liquid surface based on the thermal expansion, to calculate the total amount of change in liquid level of the liquid, and set the shift amount of the detectors **21**.

[0070] With the correction method described above, the liquid storage apparatus **10** and the semiconductor manufacturing system **1** may correct the liquid level determination positions of the liquid level sensor **20** of the container body **11** again to appropriate detectors **21**. After the correction, the controller **90** may return to step S103 of FIG. **5**, to appropriately manage the liquid of the container body **11** based on the corrected liquid level determination positions in the substrate processing of the substrate processing apparatus **50**.

[0071] The liquid storage apparatus **10**, the semiconductor manufacturing system **1**, and the correction method of the present disclosure are not limited to the first embodiment described

above, and may include various modifications. For example, the semiconductor manufacturing system **1** may recognize the amount of liquid used for each substrate processing, to adjust the timing for replenishing the liquid in the container body **11** from the replenishment unit **15**, or the supply amount (replenishment amount). For example, the semiconductor manufacturing system **1** may not perform the replenishment because of concerns that the temperature of the liquid may drop due to the replenishment while the substrate processing is being continuously performed. In this case, the controller **90** may set, in advance, the liquid level before the substrate processing to be high (increase the replenishment amount of liquid replenished from the replenishment unit **15**), and set the middle position MP to be low, thereby reducing the number of times of the replenishment of the liquid.

[0072] Further, without being limited to the configuration where the liquid in the container body **11** is heated by the heating unit **19**, the liquid storage apparatus **10** may be configured to lower the temperature of the liquid by circulating a coolant in the flow path formed in the container body **11**, and a chiller. Even when the liquid level changes due to the drop of the temperature of the liquid, the liquid storage apparatus **10** may adjust the liquid level determination positions, to appropriately respond to the change in liquid level.

[0073] FIG. **7** is a view illustrating a semiconductor manufacturing system **1A** according to a second embodiment. The semiconductor manufacturing system **1A** according to the second embodiment is different from the semiconductor manufacturing system **1** described above in that a processing gas vaporized from the liquid of the liquid storage apparatus **10** is supplied to the substrate processing apparatus **50** through the supply network **30** to perform the substrate processing on the substrate W. The substrate processing includes, for example, a deposition process by an atomic layer deposition (ALD) or a molecular layer deposition (MLD). Hereinafter, the semiconductor manufacturing system **1** that performs the ALD is described.

[0074] As in the first embodiment, the semiconductor manufacturing system **1A** includes, for example, the liquid storage apparatus **10**, the supply network **30**, and the substrate processing apparatus **50**. However, the outlet port **13** of the liquid storage apparatus **10** is attached to the ceiling of the container body **11**, to transfer the processing gas vaporized from the liquid to the supply path **31** of the supply network **30**. As for the liquid supplied to the substrate processing apparatus **50**, an appropriate liquid may be used according to the contents of the substrate processing. For example, when a deposition process for depositing a silicon nitride film is performed, an applicable liquid may be tris(dimethylamino) silane $\text{SiH}(\text{N}(\text{CH}_3)_2)_3$, so-called 3DMAS.

[0075] The supply network **30** may include a carrier gas supply unit **35** that may supply a carrier gas to the storage space **11s** of the container body **11**. The carrier gas supply unit **35** may supply the carrier gas to the container body **11**, to transfer the vaporized liquid together with the carrier gas from the container body **11** to the supply network **30**. As the carrier gas, an inert gas such as N_2 gas or noble gas may be applied.

[0076] For example, the carrier gas supply unit **35** includes an inflow path **36** through which the carrier gas may be introduced into the container body **11**, a carrier gas source **37** provided on the inflow path **36**, an opening/closing valve **38**, and a flow rate controller **39**. By controlling the opening/closing valve **38** and the flow rate controller **39**, the controller **90** supplies an appropriate amount of carrier gas from the carrier gas source **37** to the container body **11**, and transfers the processing gas vaporized from the liquid by the carrier gas.

[0077] The substrate processing apparatus **50** includes a shower head (not illustrated) or the like, for example, at the top of the processing container **51**, and substantially evenly distributes and supplies the processing gas supplied from the supply network **30** to the substrate W. As a result, a film based on the supplied processing gas is deposited on the surface of the substrate W. The substrate processing apparatus **50** may be configured to perform a plasma processing for generating a plasma in the processing gas.

[0078] In the semiconductor manufacturing system **1A** of the second embodiment as well, the liquid level sensor **20** detects the liquid level of the liquid stored in the liquid storage apparatus **10** to manage, for example, the vaporization, supply, and replenishment of the liquid, as in the first embodiment. In particular, the liquid level sensor **20** includes a plurality of (four or more) detectors **21**, and the controller **90** sets three or more liquid level determination positions for the detectors **21**, respectively, and may adjust the liquid level determination positions. As a result, the semiconductor manufacturing system **1A** may stably perform, for example, the supply of the liquid (processing gas) of the container body **11** and the replenishment of the liquid in the container body **11**

[0079] The technical concepts and effects of the present disclosure described above in the embodiments are described below.

[0080] According to a first aspect of the present disclosure, a liquid storage apparatus **10** includes: a container body **11** storing a processing liquid for a substrate processing apparatus **50** therein; a liquid level sensor **20** provided in the container body **11** and detecting a liquid level of the stored processing liquid; a replenishment unit **15** replenishing the processing liquid in the container body **11**; and a control unit (controller **90**) acquiring detection information from the liquid level sensor **20** to perform a processing. The liquid level sensor **20** includes four or more detectors **21** capable of detecting the liquid level of the processing liquid along a height direction of the container body **11**, and the controller **90** sets at least three liquid level determination positions corresponding to different detectors **21**, respectively, among the four or more detectors **21**, and controls a replenishment of the processing liquid in the container body **11** from the replenishment unit **15** based on detection information of the at least three set liquid level determination positions.

[0081] According to the configuration above, the liquid storage apparatus **10** may manage the liquid level of the processing liquid in the container body **11** with high precision, by using the liquid level sensor **20** including the four or more detectors **21**. That is, the liquid storage apparatus **10** may detect the liquid levels at various height positions in the container body **11**, by using the four or more detectors **21**. Further, the liquid storage apparatus **10** may appropriately control, for example, the timings for the discharge of the processing liquid from the container body **11** (including the vaporization of the processing liquid) and the replenishment of the processing liquid in the container body **11**, by making the at least three liquid level determination positions correspond to the detectors **21**, respectively. Further, the liquid storage apparatus **10** may change the at least three liquid level determination positions for the four or more detectors **21**, so that it is possible to set optimized liquid level determination positions according to the contents of the substrate processing or the state of the processing liquid in the container body **11**.

[0082] The at least three liquid level determination positions include a middle liquid level determination position (middle position MP) sandwiched between the uppermost liquid level determination position (highest position HP) and the lowermost liquid level determination position (lowest position LP), and the control unit (controller **90**) replenishes the processing liquid in the container body **11** from the replenishment unit **15** when the processing liquid falls below the middle liquid level determination position. As a result, the liquid storage apparatus **10** may easily perform the replenishment of the processing liquid in the container body **11**.

[0083] Further, the control unit (controller **90**) adjusts the middle liquid level determination position (middle position MP) based on the substrate processing of the substrate processing apparatus **50**. As a result, the liquid storage apparatus **10** may stably supply the processing liquid in the amount corresponding to the substrate processing.

[0084] When the substrate processing is continuously performed, the control unit (controller **90**) sets the middle liquid level determination position (middle position MP) to be lower than the preset position, and increases the replenishment amount of processing liquid replenished in the container body **11** from the replenishment unit **15**. As a result, the liquid storage apparatus **10** may continuously perform the substrate processing without replenishing the processing liquid in the container body **11** in the middle of the substrate processing, which may improve the stabilization of

the substrate processing.

[0085] The heating unit **19** is provided to heat the processing liquid stored in the container body **11**, and the control unit (controller **90**) adjusts the liquid level determination positions based on the target temperature of the processing liquid and the thermal expansion coefficient of the processing liquid. As a result, even when the thermal expansion occurs in the processing liquid due to the heating by the heating unit **19**, the liquid storage apparatus **10** may appropriately manage the amount of processing liquid in the container body **11** by changing the liquid level determination positions.

[0086] The processing liquid contains a precipitate, and the control unit (controller **90**) estimates the amount of precipitate to adjust the liquid level determination positions. As a result, even when a precipitate is deposited in the container body **11**, the liquid storage apparatus **10** may appropriately manage the amount of processing liquid in the container body **11** by changing the liquid level determination positions.

[0087] The control unit (controller **90**) estimates the amount of precipitate based on the cumulative replenishment amount of the processing liquid replenished in the container body **11**. As a result, the liquid storage apparatus **10** may estimate the amount of precipitate deposited in the container body **11** with high precision.

[0088] Based on the estimated amount of precipitate, the control unit (controller **90**) notifies a timing for replacing the container body **11**. As a result, the liquid storage apparatus **10** may request the user to replace the container body **11** at an appropriate timing.

[0089] When the processing liquid in the container body **11** is replenished from the replenishment unit **15**, the control unit (controller **90**) changes the replenishment amount of processing liquid as time elapses, based on the detection information from the liquid level sensor **20**. By changing the replenishment amount of processing liquid as time elapses, the liquid storage apparatus **10** may perform the replenishment of the liquid in a short time, and easily stabilize the liquid surface in the container body **11**.

[0090] According to a second aspect of the present disclosure, a semiconductor manufacturing system **1** includes a liquid storage apparatus **10** that stores a processing liquid and a substrate processing apparatus **50** that performs a substrate processing on a substrate **W** using the processing liquid. The liquid storage apparatus **10** includes: a container body **11** storing the processing liquid therein; a liquid level sensor **20** provided in the container body **11**, and detecting a liquid level of the stored processing liquid; a replenishment unit **15** replenishing the processing liquid in the container body **11**; and a control unit (controller **90**) acquiring detection information from the liquid level sensor **20** to perform a processing. The liquid level sensor **20** includes four or more detectors **21** capable of detecting the liquid level of the processing liquid along a height direction of the container body **11**, and the control unit sets at least three liquid level determination positions corresponding to different detectors **21**, respectively, among the four or more detectors **21**, and controls a replenishment of the processing liquid in the container body **11** from the replenishment unit **15** based on the set at least three liquid level determination positions. As a result, the semiconductor manufacturing system **1** may control the liquid level of the processing liquid in the container body **11** with high precision.

[0091] According to a third aspect of the present disclosure, a liquid replenishing method includes: preparing a liquid storage apparatus **10** storing a processing liquid for a substrate processing apparatus **50** including a container body **11** storing the processing liquid therein, a liquid level sensor **20** provided in the container body **11** and detecting a liquid level of the stored processing liquid, and a replenishment unit **15** replenishing the processing liquid in the container body **11**; setting at least three liquid level determination positions corresponding to different detectors **21**, respectively, among four or more detectors **21** of the liquid level sensor **20** that are capable of detecting the liquid level of the processing liquid along a height direction of the container body **11**; and replenishing the processing liquid in the container body **11** from the replenishment unit **15**

based on detection information of the set at least three liquid level determination positions. In this case as well, the liquid replenishing method may control the liquid level of the processing liquid in the container body **11** with high precision.

[0092] The liquid storage apparatus **10**, the semiconductor manufacturing system **1**, and the replenishment method according to the embodiments described herein are merely examples in all aspects, and are not limited.

[0093] According to an aspect, the liquid level of a processing liquid stored in a container body may be controlled with high precision.

[0094] From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A liquid storage apparatus comprising: a container configured to store a processing liquid for a substrate processing apparatus therein; a liquid level sensor provided in the container, and configured to detect a liquid level of the processing liquid stored in the container, a replenishment unit configured to replenish the processing liquid in the container; and a controller configured to acquire detection information from the liquid level sensor, and perform a processing, wherein the liquid level sensor includes four or more detectors capable of detecting the liquid level of the processing liquid along a height direction of the container, and the controller sets at least three liquid level determination positions corresponding to different detectors, respectively, among the four or more detectors, and controls a replenishment of the processing liquid from the replenishment unit to the container based on detection information of the set at least three liquid level determination positions.
2. The liquid storage apparatus according to claim 1, wherein the at least three liquid level determination positions include a middle liquid level determination position sandwiched between an uppermost liquid level determination position and a lowermost liquid level determination position, and when the processing liquid falls below the middle liquid level determination position, the controller replenishes the processing liquid in the container from the replenishment unit.
3. The liquid storage apparatus according to claim 2, wherein the controller adjusts the middle liquid level determination position based on a substrate processing by the substrate processing apparatus.
4. The liquid storage apparatus according to claim 3, wherein when the substrate processing is performed continuously, the controller sets the middle liquid level determination position to be lower than a preset position, and increases a replenishment amount of the processing liquid to be replenished in the container from the replenishment unit.
5. The liquid storage apparatus according to claim 1, further comprising: a heater configured to heat the processing liquid stored in the container, wherein the controller adjusts the liquid level determination positions based on a target temperature of the processing liquid and a thermal expansion coefficient of the processing liquid.
6. The liquid storage apparatus according to claim 1, wherein the processing liquid contains a precipitate, and the controller adjusts the liquid level determination positions by estimating an amount of the precipitate.
7. The liquid storage apparatus according to claim 6, wherein the controller estimates the amount of the precipitate based on a cumulative replenishment amount of the processing liquid replenished in the container.
8. The liquid storage apparatus according to claim 6, wherein the controller notifies a timing for

replacing the container based on the estimated amount of the precipitate.

9. The liquid storage apparatus according to claim 1, wherein when the processing liquid is replenished in the container from the replenishment unit, the controller changes a replenishment amount of the processing liquid as time elapses, based on the detection information from the liquid level sensor.

10. A semiconductor manufacturing system comprising: a liquid storage apparatus that stores a processing liquid; and a substrate processing apparatus that performs a substrate processing on a substrate using the processing liquid; wherein the liquid storage apparatus includes a container configured to store the processing liquid therein, a liquid level sensor provided in the container, and configured to detect a liquid level of the processing liquid stored in the container; a replenishment unit including a valve configured to replenish the processing liquid in the container; and a controller configured to acquire detection information from the liquid level sensor, and perform a processing, the liquid level sensor includes four or more detectors capable of detecting the liquid level of the processing liquid along a height direction of the container, and the controller sets at least three liquid level determination positions corresponding to different detectors, respectively, among the four or more detectors, and controls a replenishment of the processing liquid in the container from the replenishment unit based on the set at least three liquid level determination positions.

11. A liquid replenishing method comprising: providing a liquid storage apparatus that stores a processing liquid for a substrate processing apparatus including a container configured to store the processing liquid therein, a liquid level sensor provided in the container, and configured to detect a liquid level of the processing liquid stored in the container, and a replenishment unit including a valve configured to replenish the processing liquid in the container; setting at least three liquid level determination positions corresponding to different detectors, respectively, among four or more detectors of the liquid level sensor that are capable of detecting the liquid level of the processing liquid along a height direction of the container; and replenishing the processing liquid in the container from the replenishment unit based on detection information of the at least three liquid level determination positions set in the setting.
